Graphs — Direct proportion · Real-life graphs · Distance–time · Rates of change

You will need squared paper for the questions that say *draw*. ★ = challenge question.

A · DIRECT PROPORTION AND REAL-LIFE GRAPHS

1 The table shows the cost of apples at a market: 2 kg cost €5, 4 kg cost €10, 7 kg cost €17.50.

- **a** Are cost and mass in direct proportion? Explain how you know.
- **b** Draw a graph of cost against mass for 0 to 8 kg.
- **c** Use your graph to find the cost of 11 kg.
- **d** How many kilograms can you buy for €40?

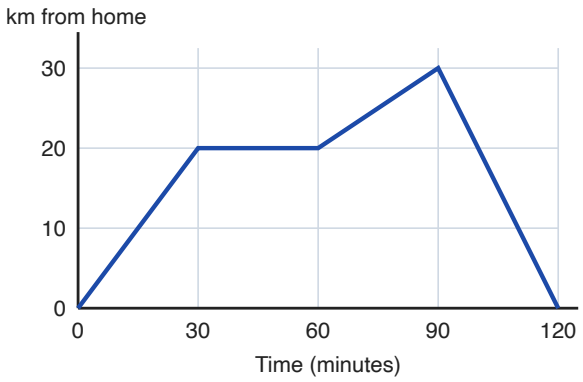
2 An electrician charges a call-out fee plus an hourly rate. Some charges: 2 hours → €70, 4 hours → €100, 5 hours → €115.

- **a** Draw the graph (time on the horizontal axis).
- **b** What is the hourly rate?
- **c** What is the call-out fee?
- **d** Are time and cost in direct proportion? Explain.

3 Four sketch graphs: **A** a straight line through the origin; **B** a curve getting steeper; **C** a straight line that starts above the origin; **D** a horizontal line. Which shows direct proportion? Explain why the others do not.

B · DISTANCE–TIME GRAPHS

4 The graph shows Nadia's cycle ride from home.



a For how long did Nadia stop?

b How far from home was she at 90 minutes?

c Which part of the ride was fastest? How do you know?

d Work out her average speed on the way home, in km/h.

e What total distance did she cycle?

C · RATES OF CHANGE AND MISLEADING GRAPHS

5 A plant is 12 cm tall and grows 1.5 cm every week.

a Write a formula for its height h cm after w weeks.

b How tall is it after 8 weeks?

c After how many weeks is it 30 cm tall?

6 A shop's poster shows a bar chart of this year's sales (480) next to last year's (450). The vertical axis starts at 440 instead of 0, and the bars have no units. Give two reasons why the chart is misleading.

7 Mobile plan A costs €10 a month plus 5 cents per MB of data. Plan B has no monthly fee and costs 10 cents per MB.

a Draw both cost graphs on the same axes for 0 to 300 MB.

b For what amount of data do the plans cost the same?

c Which plan is cheaper for 150 MB, and by how much?

8 Water freezes at $0\,°\text{C} = 32\,°\text{F}$ and boils at $100\,°\text{C} = 212\,°\text{F}$.

- **a** Draw a conversion graph from °C to °F.
- **b** Work out the gradient. What does it tell you?
- **c** Are °C and °F in direct proportion? Explain.

Check: direct proportion = straight line through the origin · gradient = rate (per hour, per kg, per °C) · on a distance–time graph, steeper means faster and flat means stopped · average speed = distance ÷ time.